

Predicting The Death of US Tourism

Final Report

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Abstract

We investigate the ability to use machine learning to predict the number of monthly tourists visiting the United States based on socio-political and economic indicators. Our target label was the number of I-94 visas granted for tourists traveling to the U.S. To accomplish this, datasets of U.S. I-94 border crossings, annual GDP, political party control and armed conflicts were chosen for feature engineering. Linear regression and a mean predictor that predicted the average monthly value were chosen as the baseline models. Through extensive feature engineering and selection, it was determined that the features with the highest impact were time-lagged arrivals, rolling averages and transformed seasonal metrics to accommodate cyclical patterns of travel. Special considerations were made to evaluate the impact of COVID on the models, as well as control data leakage. Random Forest and XGBoost were the models chosen because of their ability to handle non-linear relationships, time series, and reduce overfitting on the datasets. After feature engineering and hyperparameter tuning, an un-tuned XGBoost was found to perform the best when based on long-term arrivals data with an RMSE of 557,709 and an R^2 of 0.7530.

Introduction

In 2025, an analysis of National Travel and Tourism Office (NTTO) data showed four million fewer international visitors to the United States than in 2024—the first year-over-year decline since the COVID-19 pandemic recovery. This drop has raised concerns about economic impact, particularly as the US prepares to host the 2026 World Cup and 2028 Summer Olympics. The Congressional Research Service has flagged the trend as a risk to both visitors and participating teams, citing current immigration policy as a likely driver. The World Travel & Tourism Council projected the U.S. would lose \$12.5 billion in international visitor spending in 2025, the only one of 184 economies it analyzed to see a decline.

We want to predict the factors which are most likely to contribute to declines in U.S. inbound tourism, and determine if we can reliably predict such a decline be detected over time. Is this decline driven by home-country economic conditions, or by geopolitical and policy factors in the U.S.?

Related Work

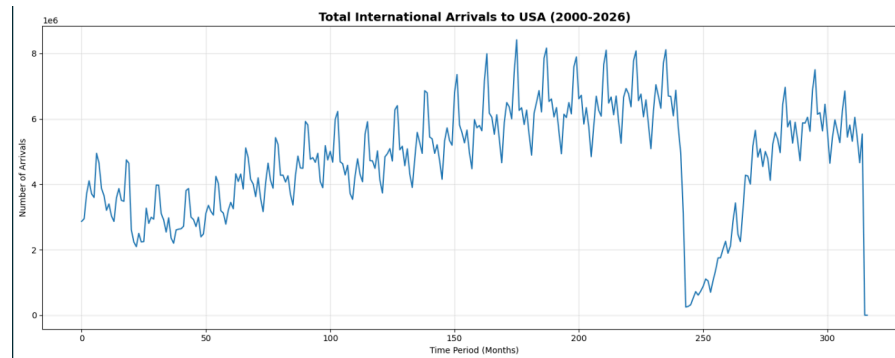
Núñez et al. (2024) systematically reviewed 397 studies on machine learning in tourism, finding that supervised and deep-learning methods dominate demand-forecasting and sentiment tasks. Liu, Tseng & Tseng (2018) applied a Vector Autoregression model to forecast destination arrivals using big data. We build on this by combining arrival forecasting with feature-importance analysis, to isolate political and geopolitical drivers. Akamavi et al. (2022) further show that security threats and instability measurably depress global travel and tourism performance, supporting our focus on the impact on U.S. tourism.

Data

Primary Target Dataset: [NTTO's I-94 arrivals program, available from 2000 to present](#)

This dataset involves the total number of I-94 visas granted for visitors to the U.S., from 2000-2026. The arrivals are separated by country and are given by a month-year format. To process, we began by dropping nulls and replacing any special characters. Then, we re-formatted the data columns to be the same formatting to enable time series processing.

The total number of arrivals for each month and year was the primary target value for the dataset. We therefore totaled the number of visas from each country for each month, and the results were used as labels for both training and test datasets. The totals are displayed in the plot below. Based on the figure, it is clear that the arrivals follow a seasonal pattern with a steep drop around the COVID-19 pandemic. Our goal with our model is to accurately predict and anticipate these trends using the features outlined below.



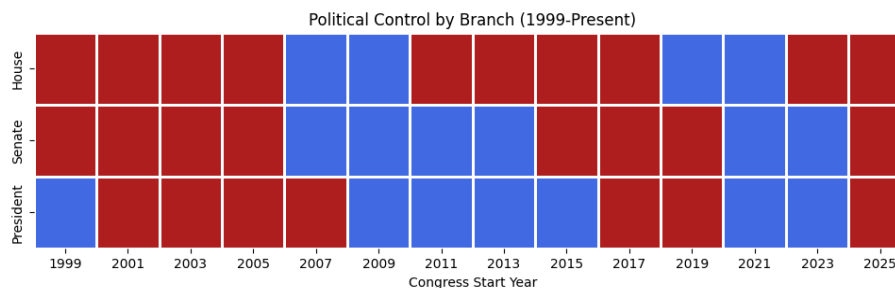
Feature Dataset #1: [Tourism Statistics Database](#) (Inbound Tourism (1) and GDP (2))

Our first feature dataset focused on the “Inbound Tourist” dataset as well as the “GDP by Country” datasets. First, we loaded in and validated the data. We then created a new dataframe with only the columns that were important to our topic. There was concern specifically with regions of origin, year, and the value or amount fields that were aligned to tourism/GDP in the United States. We removed aggregate total fields so as to not duplicate the counts. We then multiplied the values column by the “unit” column that presented as “thousands” and “percent” to have a valid value field to process.



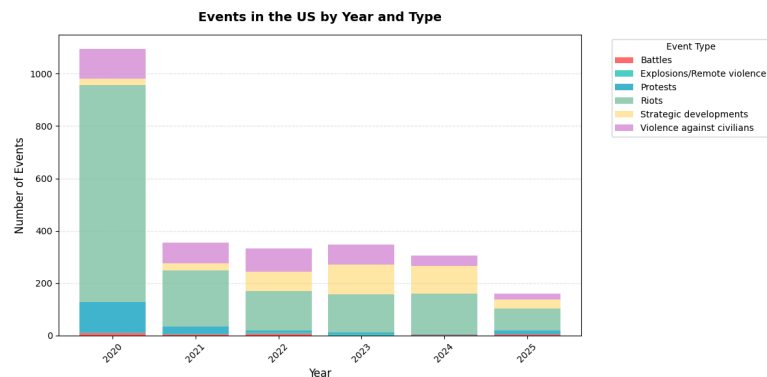
Feature Dataset #2: [Historic US Political Party Control \(1857-2025\)](#)

Our second feature dataset contains the political party in control of the U.S. House of Representatives, Senate, and Presidency. For preprocessing, we decided to only include data from the years 2000 and onwards to mirror the information available from I-94 arrivals. Using a heatmap, we can visualize the changes in control over time, which can be used to derive different features.



Feature Dataset #3: [Armed Conflict Location & Event Data \(ACLED\) \(in the U.S., 2020–Present\)](#)

Our last feature dataset includes records of significant events from multiple countries. We kept only the rows for the U.S., from 2020 onward (no armed conflict data was recorded for the U.S. prior to 2020). We cross-checked the filtered data against events, i.e., George Floyd protests and the Jan. 6 Capitol riot, and chose to include only politically-violent events, to allow for the model to utilize context-driven features.



Dataset Challenges and Implications: Because we are utilizing datasets from various sources, the time periods rarely line up. Additionally, features pulled from either the political party or ACLED datasets often lack critical context, which may be difficult for a model to pull. This includes impactful policies passed during an administration that could affect travel, or particularly impactful protests or violent conflicts which lead to travel bans or negative sentiment. Finally, COVID-19 presented a challenge for training our model. Between omitting the information and keeping it for our features, we decided to use the information from the pandemic to train our model in our attempt to provide robustness to significant outliers.

Methods

Overview: The goal of our model was to determine the best features within our datasets and predict the projected number of tourists to the U.S. in our test dataset. The I-94 data will provide the number of arrivals at a given time as our “ground truth” metric for evaluation, which will indicate whether our experiments improve or degrade the model. We needed to ensure an appropriate training/validation/test split was done on our time-series data. Because a random split may result in leakage, we used a pre-determined ‘split-date’ of January 1st, 2022 to divide the training and test data. This was applied to both our features and label. Additionally, our model selection needed to reflect our reliance on features to contextualize time-based events such as COVID-19, politically-violent conflicts, and political change.

Baseline Models: A simple mean-value predictor and linear regression model were used as our baselines. The mean-value predictor represented a straightforward performance metric to evaluate our improvements on. Linear regression served as the first-step in feature engineering by evaluating the coefficients. Outliers were quickly detected and investigated as we fine-tuned our features.

Improving on Baseline Performance: For comparison, we created models generated by both a Random Forest and Gradient Boosted decision tree model. Decision trees provide several advantages over our baselines. Features with non-linear relationships to our outcome will be reflected more appropriately in a decision tree. Additionally, they have greater robustness to outliers and unusual events. Experimentation to improve performance focused on comparing ensemble learning performance between Random Forest and Gradient Boosting. A Random Forest model allowed us to quickly determine whether or not decision tree-based learning creates tangible improvements over our baseline results. From there, we determined if hyperparameter tuning via boosting would result in a noticeable increase in accuracy.

Experiment, Results and Discussion

Feature Engineering: Features from all datasets were engineered to enable fine tuning of the models. Due to the data being time-series impacted by seasonality, there were issues found with data leakage. To prevent this, the features were lagged and averaged over a 3 month rolling period to represent the time between booking and arriving. Arrivals were also transformed by sine and cosine to accommodate for annual cycles, like peak summer periods. The data was then broken into periods—pre-COVID, COVID, and post-COVID—to understand the impact of the pandemic on all of the features and the models themselves. Our full set of pre-engineering and final features can be found in the appendix.

Results and Discussion: The results of our baseline models, engineered features selected, and the pre-tuned/tuned Random Forest and XGboosting were as follows:

Mean-Value Predictor Baseline

RMSE: 1,125,259

MAE: 841,561

R²: 0.0

Linear Regression Baseline

RMSE : 60,160,974

MAE : 50,218,580

R² : -2872.7939

Pre-Tuned Random Forest

RMSE : 594,103

MAE : 460,445

R² : 0.7197

Pre-Tuned XGBoost (Best Performance)

RMSE : **557,709**

MAE : **406,561**

R² : **0.7530**

Linear regression's accuracy metrics were heavily skewed by COVID-19 and ACLED features in the training data, therefore the mean-value predictor served as our primary performance baseline. Though the results are similar, we found that the XGBoost model out-performed the Random Forest model by a small margin. Hyperparameter tuning was then done using RandomizedSearchCV with TimeSeriesSplit to preserve chronological order and the tuning was only done on training data. The results were as follows:

Tuned Random Forest

RMSE: 594,103 -> 780,255

MAE: 460,445 -> 670,333

R² : 0.7197 -> 0.5166

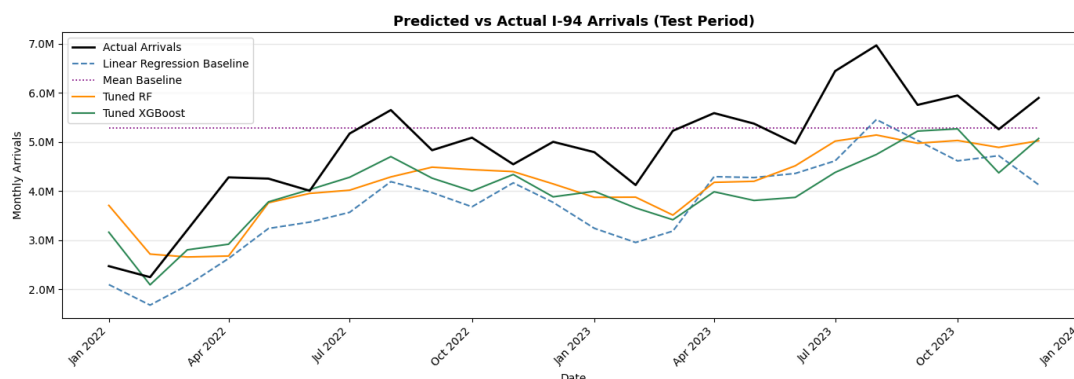
Tuned XGBoost

RMSE : 557,709 -> 581,359

MAE : 460,561 -> 499,689

R² : 0.7530 -> 0.7316

Tuning exposed the difficulty our decision tree models had in generalizing the patterns it observed in the training data. This comes as no surprise, as our time-based features outperformed all of the political/socio-economic features derived from our feature data. Additionally, our models consistently underpredicted the actual number of arrivals due to decision tree's built-in hesitation to predict greater values than seen in the training data. More detailed plots depicting XGBoost performance and residuals can be found in the appendix.



Conclusion

Feature engineering resulted in greater performance increases than hyperparameter tuning, while introduction of time-based features yielded the best results. Our initial features yielded a pre-tuned Random Forest model with an RMSE of greater than 3.5 million and an R^2 of -23.73, whereas the model with our final features had an RSME of ~594,000, and an R^2 equal to 0.72. Time-based features, such as lag and rolling values pulled from previous months in the I-94 data, yielded the most consistent improvements in accuracy. Our design decision to include the COVID-19 pandemic in our training data resulted in a significant outlier which our feature engineering could not overcome. Further experimentation on this problem would need to focus on including seasonal models specializing in forecasting such as SARIMA to fully account for the cyclic nature of tourism. A centralized dataset where features share the same timeframe will further enhance the models ability to make predictions with greater context and accuracy. Although this is a complex problem with a multitude of potential confounding variables, these types of experiments only serve to hopefully further our understanding of travel and tourism in the U.S.

Contributions

Peter Huang: I-94 EDA and data cleaning, initial feature engineering, baseline and initial RF and XGBoost modeling

Abigail Kreutz: ACLED EDA and data cleaning, ACLED feature engineering, and secondary RF and XGBoost modeling including ACLED event and time-series features

Hope Tadsen: GIT repo creation, inbound tourism and GDP EDA and data cleaning, visual revisions, deck consolidation, and report draft

Github repository: https://github.com/UC-Berkeley-I-School/Summer06_207_FinalProject_Group_APH

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Appendix

Feature Engineering Timeline

Initial Features

Congressional

is_rep_house, is_rep_senate, is_rep_presidentis_unified, house_changed, senate_changed, pres_changed, is_election_year

Tourism

tourism_value, tourism_pct_change, gdp_value, gdp_pct_change

Politically-Violent Events

acled_incidents, acled_incidents, acled_fatal_events, acled_total_fatalities, acled_any_event, acled_any_fatal_event, acled_high_activity_flag, acled_events_lag_1, acled_fatal_events_lag_1, acled_events_lag_2, acled_fatal_events_lag_2, acled_events_roll_3, acled_fatalities_roll_3, acled_states, acled_event_types

Intermediate Features

Congressional

is_rep_house, is_rep_senate, is_rep_president, is_unified, house_changed, senate_changed, pres_changed, is_election_year

Tourism

tourism_value, tourism_pct_change, gdp_value, gdp_pct_change

Politically-Violent Events

acled_incidents, acled_events_roll_3, acled_events_lag_1

Time-Based

arrivals_lag_1, arrivals_lag_2, arrivals_lag_12, arrivals_roll_3, month_sin, month_cos

Final Features

Politically-Violent Events

acled_incidents, acled_events_roll_3, acled_events_lag_1

Time-Based

arrivals_lag_1, arrivals_lag_12, arrivals_roll_3, month_sin, month_cos

Pre-Tuned Decision Tree Model Parameters and Results

Pre-Tuned Random Forest

Parameters:

n_estimators=500,
max_depth=3,
min_samples_leaf=2,
max_features=0.8

Results:

RMSE : 594,103

MAE : 460,445

R^2 : 0.7197

Pre-Tuned XGBoost (Best Performance)

Parameters:

n_estimators=250,
max_depth=2,
learning_rate=0.03,
min_child_weight=2,
subsample=0.8,
colsample_bytree=0.9,
reg_lambda=5

Results:

RMSE : 557,709

MAE : 406,561

R^2 : 0.7530

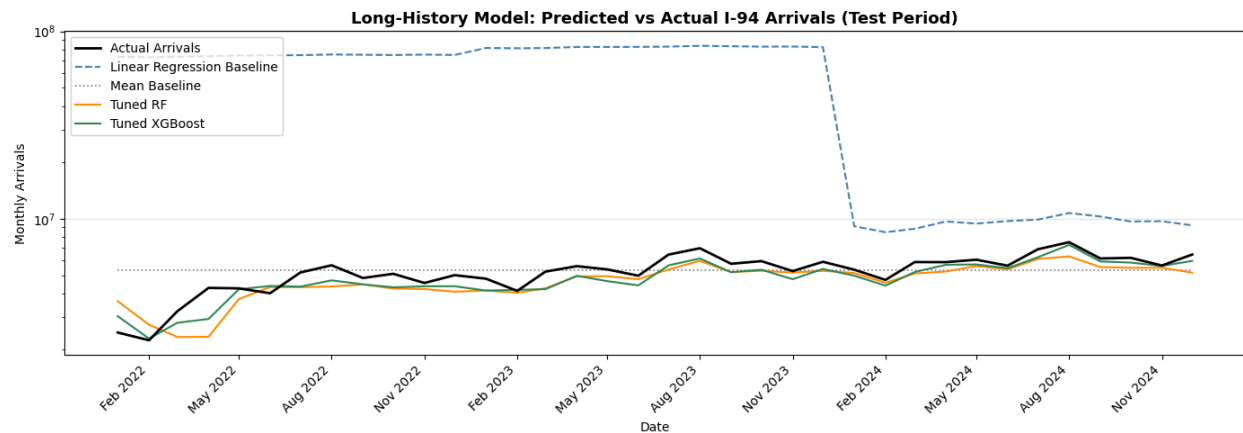


Figure: Final XGBoost/Random Forest model performance

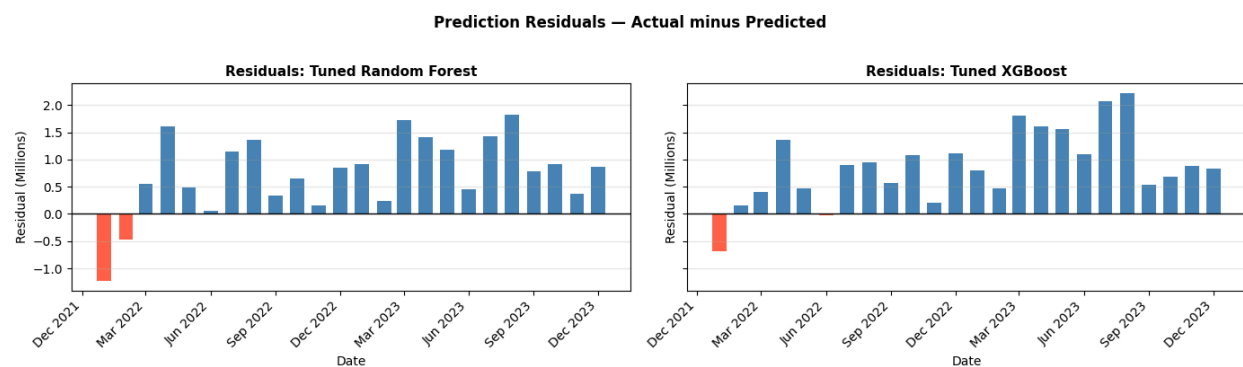


Figure: Residuals from tuned models